

Executive Overview: SchoolPilot is a next-generation, cloud-based school management platform engineered specifically for private K-12 day schools in Nigeria. Operating strictly under a **Software-Only** architectural core, SchoolPilot eliminates expensive biometric scanners, RFID readers, and specialized hardware. It turns standard smartphones, browsers, and existing office PCs into an integrated hub for administration, AI-assisted teaching, parent engagement, computer-based testing (CBT), and automated fee processing.

1. Zero-Hardware & Software-Only Architecture

Traditional school management platforms require costly biometric readers, facial scanners, and smart cards. SchoolPilot completely replaces hardware procurement with standard mobile and desktop web capabilities:

- **QR Code Camera Attendance:** Teachers and gatekeepers scan student badges or mobile passes directly using the built-in smartphone camera view for real-time attendance logging.
- **GPS Geofenced Staff Clock-In:** Staff log arrival and departure via smartphone GPS verification, enforcing authentic clock-ins strictly within school premises.
- **Real-Time Bus Location Broadcasting:** Bus drivers run the mobile app during active routes, allowing parents to track bus arrival live without vehicle GPS tracker hardware.
- **Phone Camera Library Barcoding:** Library check-in and checkout uses the phone camera to scan standard book barcodes, removing the need for handheld barcode guns.
- **Printable Student & Staff ID Cards:** Generates high-resolution, print-ready PDF ID cards complete with QR codes and school branding, printable on standard office printers.
- **Resilient Offline Desktop CBT:** Lightweight desktop app for CBT exams that runs without internet during outages and syncs scores automatically when reconnected.

2. Core Academic & Administrative Management (K-12 Focused)

Designed specifically around Nigerian K-12 academic structures (Sessions, Terms, Classes, Arms/Streams) rather than generic higher-education faculty concepts.

- **Nigerian Academic Structure Native:** Full support for 3-Term sessions, Continuous Assessment (CA) weightings, Termly Broadsheets, JSS/SS divisions, and national curriculum standards (WAEC, NECO, JAMB).
- **Student Information System (SIS):** Centralized digital student dossiers tracking profiles, guardian linkages, medical notes, academic history, and NDPA 2023 compliant data protection.
- **Automated Conflict-Free Timetabling:** Intelligent schedule generator balancing subject periods, teacher availability, and classroom allocations without scheduling overlaps.
- **Unified Multi-Tenant Web & Mobile App:** Modern Next.js web dashboard and Flutter cross-platform mobile app providing role-optimized access for Admins, Teachers, Students, and Parents.

3. AI-Powered Teaching & Learning Ecosystem

- **AI Teacher Copilot:** Generates comprehensive lesson plans, practice worksheets, and assignment quizzes tailored to specific term topics and curriculum goals in seconds.
- **AI Report Card Remarks with Human Oversight:** Generates personalized, constructive report card remarks for every student. AI comments stay in *pending_approval* status until reviewed and approved by the teacher.
- **24/7 AI Student Curriculum Tutor:** Interactive personal AI assistant calibrated strictly to the school's actual syllabus and teacher-assigned coursework for safe round-the-clock learning support.
- **Admin AI Risk & Revenue Insights:** Automated predictive alerts for school heads identifying students showing sudden score drops, attendance anomalies, or tuition delinquency risks.

4. Computer-Based Testing (CBT) & Assessment Engine

- **Rich Media & LaTeX Math Questions:** Supports multiple choice, true/false, and theory questions enriched with images and inline LaTeX math formulas rendered offline via bundled KaTeX.
- **Anti-Cheat & Exam Hall Controls:** Question randomization, option shuffling, strict timed countdowns, browser focus loss monitoring, and instant automated grading.
- **WAEC/JAMB Standardized Practice:** Pre-loaded question banks allowing students to simulate national standardized examinations under authentic timed exam hall conditions.

5. Fee Management & Financial Integration

- **Integrated Payment Gateways:** Native integration with Paystack and Flutterwave enabling tuition payment via Debit Card, Bank Transfer, or USSD with automated instant receipts.
- **Parent Financial Ledger & Invoicing:** Real-time itemized breakdown of tuition, uniforms, bus services, and extracurricular fees with partial payment support and clear balance tracking.
- **WhatsApp & SMS Communication Hub:** Automated payment reminders, attendance alerts, exam results, and school announcements sent directly to parents' mobile phones.

6. User Roles & System Permissions Matrix

Role	Primary Responsibilities & Access Scope	Key Features Available
Super Admin	Platform operators & system maintenance	Multi-school onboarding, platform security, subscription billing, cross-school analytics
School Admin	Principal, Proprietor, & Front Office	Staff/Student records, fee structures, timetables, broadsheet approval, AI insight dashboard
Teacher	Academic staff & Subject Instructors	Attendance scanning, CA grading, AI lesson planner, AI report card remark approval
Student	Enrolled learners (JSS1 - SS3)	CBT exam portal, assignment submission, term results view, 24/7 AI curriculum tutor
Parent	Guardians & Sponsors	Online fee payments, real-time bus tracking, child attendance, report cards, teacher chat

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